Anshu Dahal

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EDUCATION

Truman State University **Kirksville, MO**
Bachelor of Science in Computer Science and Data Science *May 2027*
President's Scholarship Recipient, President's Honors List, Advanced Level Education Scholarship GPA: 4.0 / 4.0

TECHNICAL SKILLS

Programming and Data: Python, SQL, R, pandas, scikit-learn, PyTorch, statistical modeling, data visualization | Applied AI: LLMs, RAG, LlamaIndex, Hugging Face embeddings, RAGAS, retrieval evaluation, prompt engineering | Engineering: FastAPI, REST APIs, Docker, Linux, Git/GitHub, SSH, Quarto, React, Node.js, MongoDB

PROFESSIONAL EXPERIENCE

Computational Cancer Genomics Summer Program, Washington University in St. Louis **St. Louis, MO**
Visitor Program Trainee *May 2026 - Jul 2026*

- Designed and evaluated a patient-facing RAG assistant that translated complex MyeloSeq cancer reports into grounded explanations using LlamaIndex, Hugging Face embeddings, metadata filtering, retrieval, reranking, and an OpenAI-compatible endpoint.
- Built a balanced RAGAS benchmark to identify grounding failures and compare retrieval strategies using answer relevancy, faithfulness, context precision, and context recall.
- Engineered reproducible R and Python workflows to clean, standardize, integrate, and validate large clinical, demographic, genomic, survival, and social determinants of health datasets for breast cancer research.
- Converted mutation records into gene-presence matrices, analyzed TP53 and PIK3CA patterns, and performed Kaplan-Meier overall and progression-free survival analyses.
- Delivered reproducible analyses, visualizations, and technical documentation with R and Quarto for clinical and technical stakeholders.
- Used Docker containers and remote Linux servers through SSH for reproducible development, data processing, and deployment workflows.
- Collaborated with clinical researchers and engineers through GitHub code reviews, technical discussions, and iterative development of research and AI tools.

Office of Community and Enrichment, Truman State University **Kirksville, MO**
Website Designer *Jan 2024 - Present*

- Maintain oce.truman.edu and translate requests from university stakeholders into tested, accessible website improvements for students, staff, and community partners.
- Prototype human-in-the-loop AI workflows that convert maintenance requests into reviewable actions while preserving staff approval before production changes.

PROJECTS

Student Wellness Check-In
Kotlin, Jetpack Compose, MVVM, Room, FastAPI, Qwen, Gemini API

- Built an end-to-end Android wellness application with local persistence, interactive health visualizations, and a Python FastAPI service, translating user needs into a working prototype.
- Improved LLM and REST reliability through protected API keys, model routing, safe fallbacks, and tests covering latency, memory use, failure handling, safety, and response quality.
- Used MVVM to separate interface, state, persistence, and service logic, supporting maintainable development and rapid feature iteration.
- Integrated Qwen and Gemini model workflows behind a FastAPI service while keeping interface, persistence, and AI components independently testable.

Memory Garden
React, Node.js, MongoDB, Mongoose, OpenAI API

- Built a full-stack AI journaling product with responsive React interactions, Node.js services, MongoDB schemas, interactive visualizations, and generated summaries.
- Iteratively tested accessibility, user feedback, response quality, and persistent context to improve reliability and usability.
- Designed MongoDB Atlas and Mongoose data models that preserved user records and supported maintainable application growth.

DDI Dermatology AI Fairness Research
Python, PyTorch, EfficientNet-B3, scikit-learn, Grad-CAM

- Improved dermatology model AUC for darker skin tones from 0.61 to 0.77 and reduced false-negative rate from 88% to 42% through fairness regularization, calibration, and structured error analysis.
- Engineered a reproducible PyTorch training and evaluation pipeline with 10-fold Monte Carlo validation, checkpointing, data-quality checks, and Grad-CAM debugging.
- Produced group-level AUC, error-rate, calibration, and weighted-harm analyses with eight visualization workflows to make model behavior transparent and reviewable.

ACTIVITIES

Google Developers Student Club **Kirksville, MO**
Lead *Aug 2025 - Present*

- Lead the student developer community, coordinating technical events, resources, and outreach.
- Led a 100-person hackathon and represented GDG at developer meetups and keynote sessions in Santa Clara.